



Contact Information:

EngiNet Office Staff: Janice Kinzer

Email: enginnet@binghamton.edu

Phone: 1-800-478-0718 or 607-777-4965

Fax: 1-800-258-6640 or 607-777-6254

Media Production Operator: Lisbel Feliz Garo

Instructor: Prof. Prof. Susan Lu

Email: slu@binghamton.edu

Phone: 607-777-4908

TA: Samer Abubaker

Office Hours:

SSIE 605

Applied Multivariate Data Analysis

Professor Susan Lu

Lecture 10

Agenda

□ Factor analysis

- One factor model
- Two factor model
- Extracting/Deriving Factors
- SAS and output interpretation
- examples

Factor Analysis Terminology

- Indicator or measure: the interrelated variables under study.
- Common or latent factor: underlying, unobserved, common information shared across indicator.

Factor Analysis Terminology

- **Pattern loading:** coefficients of the common factor
- **Structure loading:** Pearson correlation between indicator and latent factor
- **Communality:** the square of the pattern loading, which is also proportion of variance that is accounted for by the factor.

Variance Structure

- Total variance of any indicator includes:
 - The communality of the indicator: variance that is in common which is given by the square of the pattern loading.
 - Unique or specific or error variance: Variance of the indicator variable minus the communality.

Example -Student Performance

Correlation Matrix for One-Factor Model

	M	P	C	E	H	F
M	1.000					
P	.56	1.000				
C	.72	.63	1.000			
E	.48	.42	.54	1.000		
H	.40	.35	.45	.30	1.000	
F	.52	.46	.59	.39	.33	1.000

Correlation Matrix for one-factor Model

Example -Student Performance

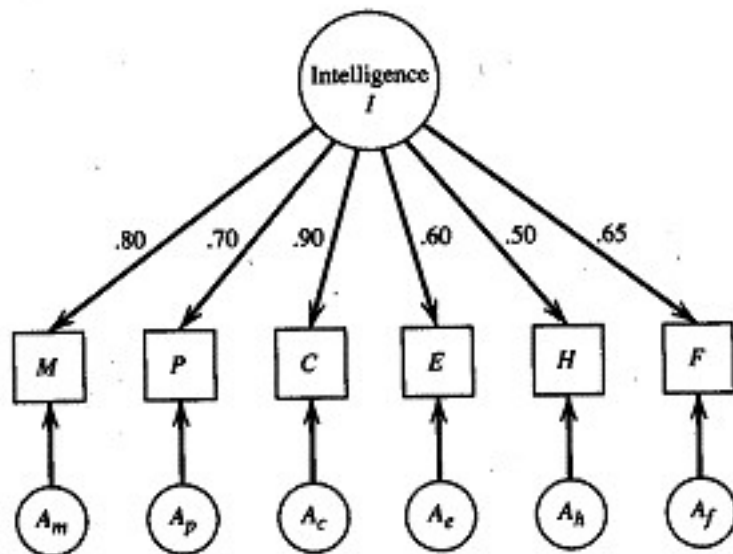


Figure 5.1 Relationship between grades and intelligence.

$$M = .80I + A_m;$$

$$C = .90I + A_c;$$

$$H = .50I + A_h;$$

$$P = .70I + A_p$$

$$E = .60I + A_e$$

$$F = .65I + A_f.$$

- The correlation between any two indicators is equal to the **products** their respective **pattern loadings**

Example -Student Performance

Table 5.1 Communalities, Pattern and Structure Loadings, and Correlation Matrix for One-Factor Model

Communalities

Variable	Communality	Error or Unique Variance	Pattern Loading	Structural Loading	Shared Variance
M	.640	.360	.800	.800	.640
P	.490	.510	.700	.700	.490
C	.810	.190	.900	.900	.810
E	.360	.640	.600	.600	.360
H	.250	.750	.500	.500	.250
F	.423	.577	.650	.650	.423
Total	2.973	3.027			2.973

Objectives of EFA

- Identify the smallest number of common factors that best explain the **correlations** among the indicators.
- Identify the most **plausible** factor solution.
- Estimate the pattern and structure loadings, communalities, and the unique variances of the indicators.
- Provide an interpretation for the common factor(s) and/or estimate the factor scores.

Two-Factor Model

- Used when single factor can't explain the interrelation enough

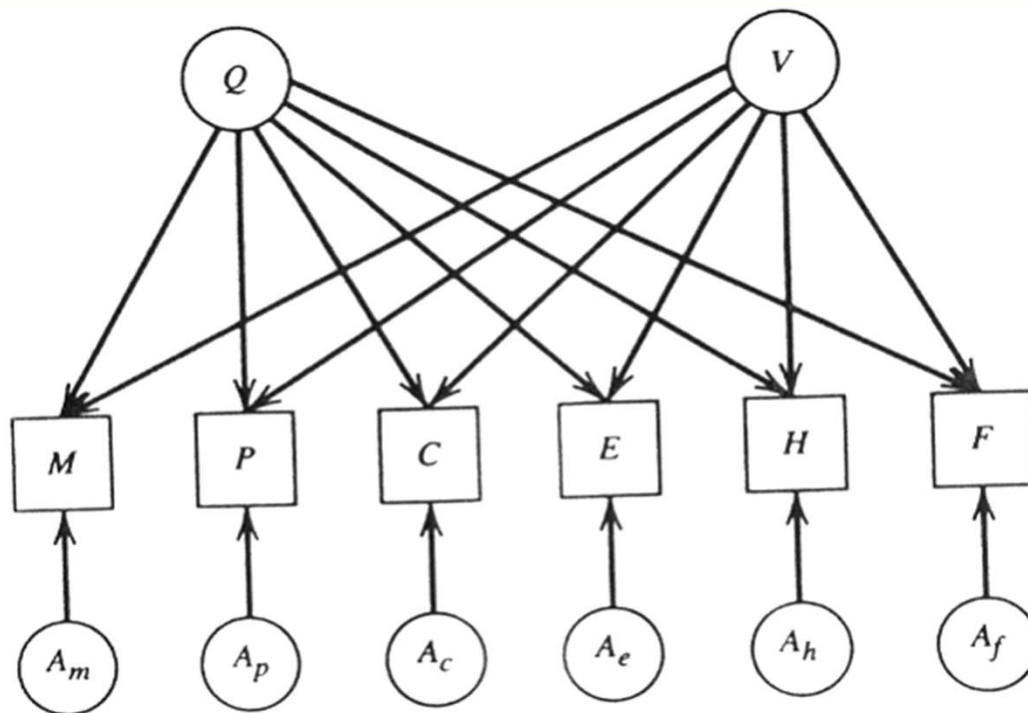


Figure 5.2 Two-factor model.

$$M = .800Q + .200V + A_m;$$

$$C = .600Q + .300V + A_c;$$

$$H = .150Q + .820V + A_h;$$

$$P = .700Q + .300V + A_p$$

$$E = .200Q + .800V + A_e$$

$$F = .250Q + .850V + A_f.$$

Observations and facts©

Pattern and Structure Loadings and Shared Variance

Variable	Pattern Loading		Structure Loading		Shared Variance	
	Q	V	Q	V	Q	V
M	.800	.200	.800	.200	.640	.040
P	.700	.300	.700	.300	.490	.090
C	.600	.300	.600	.300	.360	.090
E	.200	.800	.200	.800	.040	.640
H	.150	.820	.150	.820	.023	.672
F	.250	.850	.250	.850	.063	.723
Total					1.616	2.255

Correlation Matrix

	M	P	C	E	H	F
M	1.000					
P	.620	1.000				
C	.540	.510	1.000			
E	.320	.380	.360	1.000		
H	.284	.351	.336	.686	1.000	
F	.370	.430	.405	.730	.735	1.000

Observations and facts[©]

Communalities

Variable	Communalities		Total	Unique Variance
	Q	V		
M	.640	.040	.680	.320
P	.490	.090	.580	.420
C	.360	.090	.450	.550
E	.040	.640	.680	.320
H	.023	.672	.695	.305
F	.063	.723	.786	.214
Total	1.616	2.255	3.871	2.129

Observations and Facts

- Variance of any indicator can be decomposed into three components
- Pattern loading and structure loading is same when the factors are independent (orthogonal)
- The correlation between any two indicators is equal to the sum of the products of respective pattern loadings for each factor

Observations and Facts

- The total variance of an indicator that is in common with two latent factors, Q and V , is called the **total communality**.
- The variance that is in common with **specific factor** is 1 minus the communality.

FA with more than two factors

$$x_1 = \lambda_{11}\xi_1 + \lambda_{12}\xi_2 + \cdots + \lambda_{1m}\xi_m + \epsilon_1$$

$$x_2 = \lambda_{21}\xi_1 + \lambda_{22}\xi_2 + \cdots + \lambda_{2m}\xi_m + \epsilon_2$$

$$\vdots$$

$$x_p = \lambda_{p1}\xi_1 + \lambda_{p2}\xi_2 + \cdots + \lambda_{pm}\xi_m + \epsilon_p.$$

FA V. S. PCA

$$\xi_1 = w_{11}x_1 + w_{12}x_2 + \cdots + w_{1p}x_p,$$

Factor Indeterminacy

- FA solution is not unique
 - Factor rotation is not unique
 - Infinity of factor solutions (infinity of factor rotation)
 - Interpretable is the one we want!
 - Estimation of communality is not unique

Factor Indeterminacy©

□ Factor rotation problem – Example

Correlation Matrix

	M	P	C	E	H	F
M	1.000					
P	.620	1.000				
C	.540	.510	1.000			
E	.320	.380	.360	1.000		
H	.284	.351	.336	.686	1.000	
F	.370	.430	.405	.730	.735	1.000

Factor Indeterminacy

□ Factor rotation problem – Example

$$\begin{array}{ll} M = .800Q + .200V + A_m; & P = .700Q + .300V + A_p \\ C = .600Q + .300V + A_c; & E = .200Q + .800V + A_e \\ H = .150Q + .820V + A_h; & F = .250Q + .850V + A_f. \end{array}$$

$$\begin{array}{ll} M = .667Q - .484V + A_m; & P = .680Q - .343V + A_p \\ C = .615Q - .267V + A_c; & E = .741Q + .361V + A_e \\ H = .725Q + .412V + A_h; & F = .812Q + .355V + A_f \end{array}$$

Factor Indeterminacy©

□ Factor rotation problem – Example

Communalities

Variable	Communalities		Total	Unique Variance
	Q	V		
M	.445	.234	.679	.321
P	.462	.118	.580	.420
C	.378	.071	.449	.551
E	.549	.130	.679	.321
H	.526	.170	.696	.304
F	.659	.126	.785	.215
Total	3.019	.849	3.868	2.132

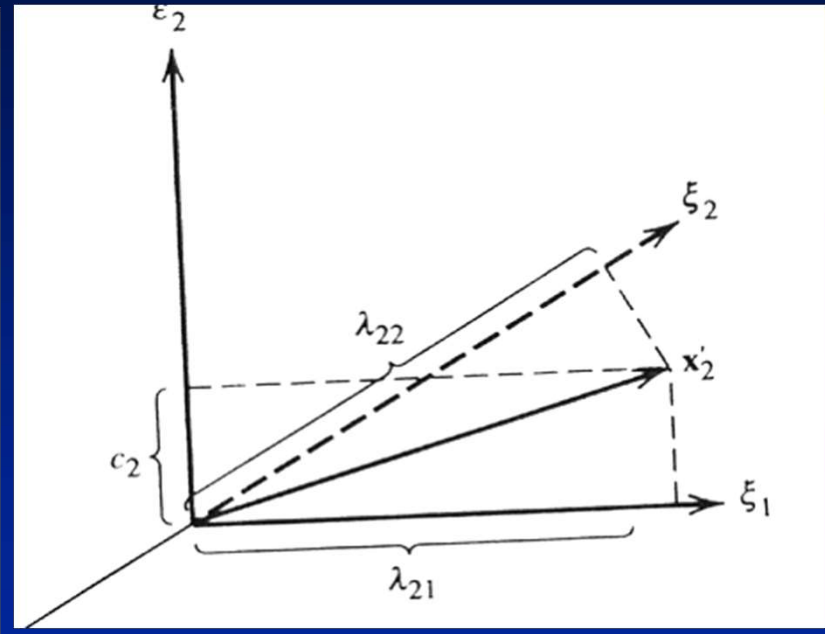
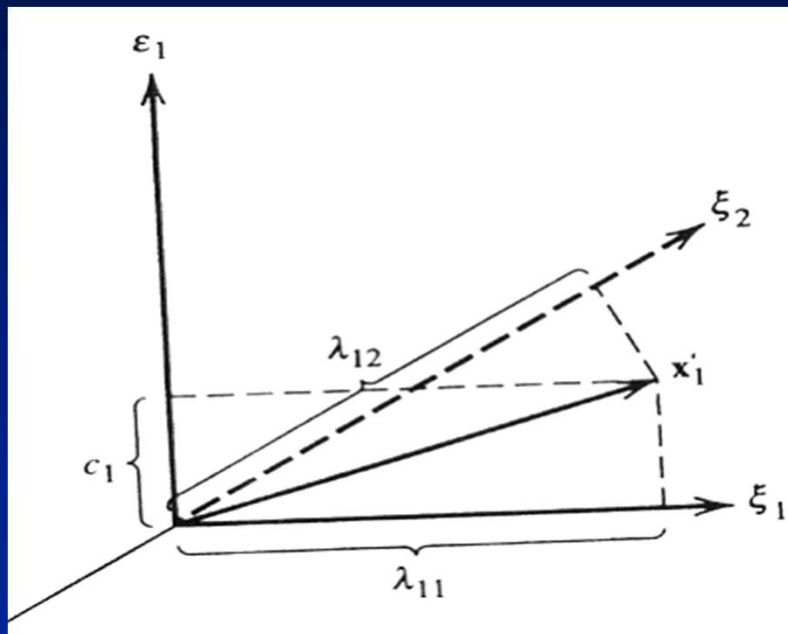
Factor Indeterminacy©

□ Factor rotation problem – Example

Pattern and Structure Loadings and Shared Variance

Variable	Pattern Loading		Structure Loading		Shared Variance	
	Q	V	Q	V	Q	V
M	.667	–.484	.667	–.484	.445	.234
P	.680	–.343	.680	–.343	.462	.118
C	.615	–.267	.615	–.267	.378	.071
E	.741	.361	.741	.361	.549	.130
H	.725	.412	.725	.412	.526	.170
F	.812	.355	.812	.355	.659	.126
Total					3.019	.849

Indeterminacy due to Estimates of Communalities



$$\|x'_1\|^2 = \lambda_{11}^2 + \lambda_{12}^2 + c_1^2$$
$$\|x'_2\|^2 = \lambda_{21}^2 + \lambda_{22}^2 + c_2^2.$$

Indeterminacy due to Factor Rotation

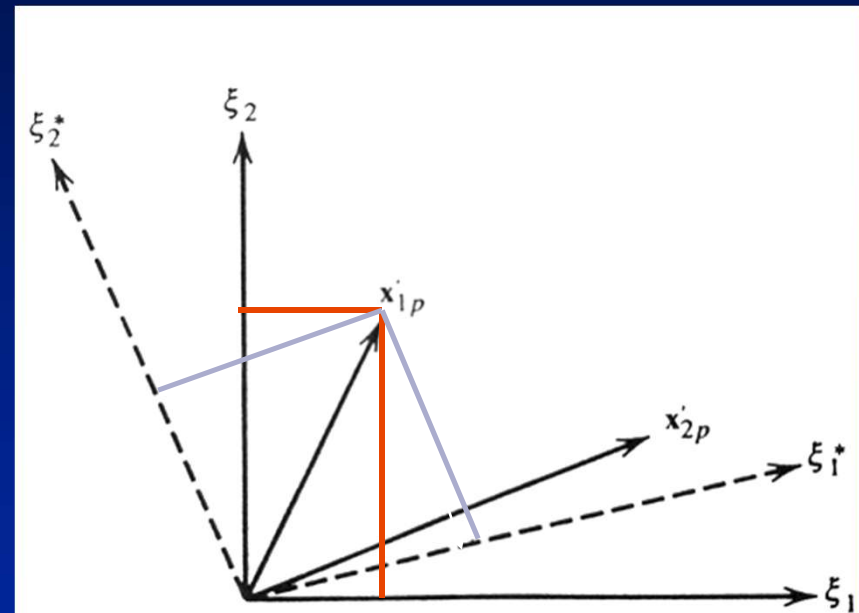
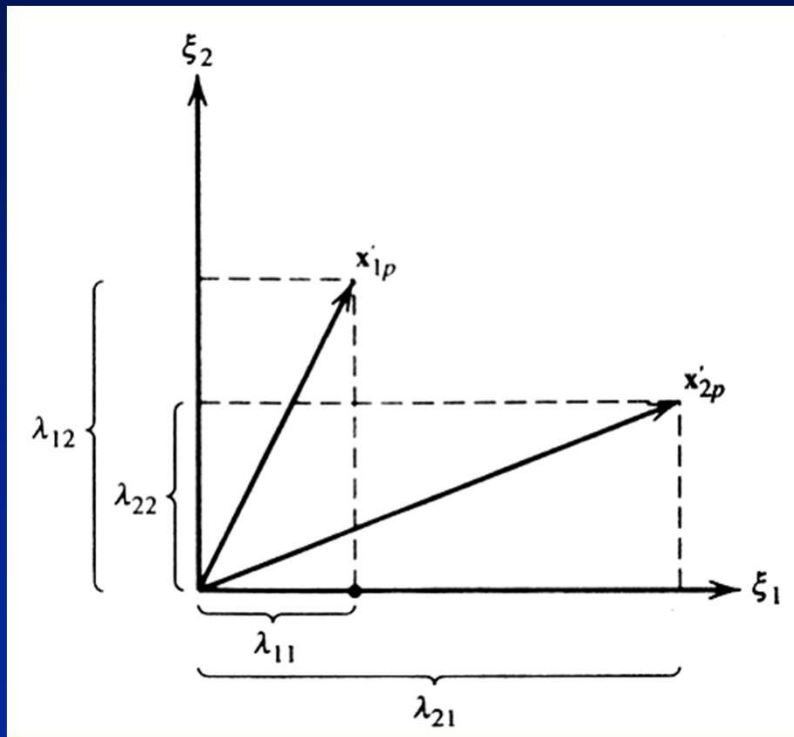
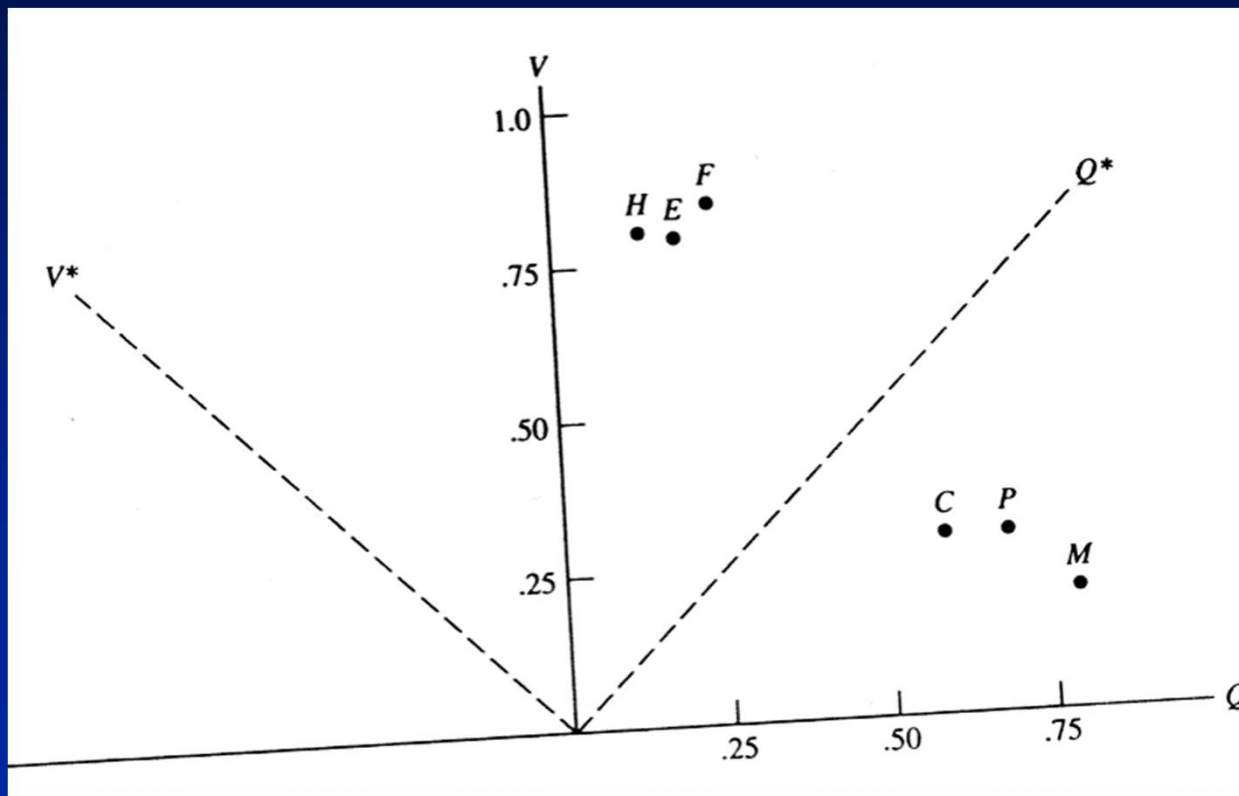


Figure 5.6 Rotation of factor solution.

Indeterminacy due to Factor Rotation©



Initial Extracting/Deriving Factors

□ Approaches

- Principal Components Factoring (PCF)
- Principal Axis Factoring (PAF)

Initial Extracting/Deriving Factors

- Principal Components Factoring (PCF)
 - Starting with PCA using the correlation matrices (Diagonal Position=?).
 - Standardizing each principle component.
 - Determining the # of PCs

Principal Components Factoring (PCF)

1	EIGENVALUE	DIFFERENCE	PROPORTION	CUMULATIVE
PRIN1	3.36689	2.17285	0.561149	0.56115
PRIN2	1.19404	0.68703	0.199007	0.76016
PRIN3	0.50701	0.13516	0.084501	0.84466
PRIN4	0.37185	0.05873	0.061974	0.90663
PRIN5	0.31312	0.06602	0.052186	0.95882
PRIN6	0.24709	.	0.041182	1.00000

2	EIGENVECTORS					
	PRIN1	PRIN2	PRIN3	PRIN4	PRIN5	PRIN6
M	0.367802	0.509824	-.266979	0.727665	0.047857	0.041663
P	0.391381	0.409168	-.485915	-.664648	-.005389	0.038775
C	0.371982	0.382542	0.831629	-.152048	-.003335	0.023552
E	0.432206	-.374995	-.021560	0.065466	-.741529	0.343453
H	0.421900	-.421447	-.002701	0.011605	0.666963	0.446543
F	0.456476	-.328759	-.023047	0.034749	0.054439	-.823921

Principal Components Factoring (PCF)

$$\begin{aligned}\xi_1 &= .368M + .391P + .372C + .432E + .422H + .456F \\ \xi_2 &= .510M + .409P + .383C - .375E - .421H - .329F \\ \xi_3 &= -.267M - .486P + .832C - .022E - .003H - .023F \\ \xi_4 &= .728M - .665P - .152C + .065E + .012H + .035F \\ \xi_5 &= .048M - .005P - .003C - .742E + .667H + .054F \\ \xi_6 &= .042M + .039P + .024C + .343E + .447H - .824F.\end{aligned}$$

Standardize PCA score

$$\frac{\xi_1}{\sqrt{3.367}} = .368M + .391P + .372C + .432E + .422H + .456F$$

Principal Components Factoring (PCF)

Standardized PCA scores

$$\xi_1 = .675M + .717P + .683C + .793E + .774H + .837F$$

$$\xi_2 = .557M + .447P + .418C - .410E - .461H - .359F$$

$$\xi_3 = -.190M - .346P + .592C - .015E - .002H - .016F$$

$$\xi_4 = .444M - .405P - .093C + .040E + .007H + .021F$$

$$\xi_5 = .027M - .093P - .002C - .415E + .373H + .030F$$

$$\xi_6 = .021M + .019P + .012C + .171E + .222H - .409F.$$

Principal Components Factoring (PCF)

Convert to FA format

$$M = .675\xi_1 + .557\xi_2 - .190\xi_3 + .444\xi_4 + .027\xi_5 + .021\xi_6$$

$$P = .717\xi_1 + .447\xi_2 - .346\xi_3 - .405\xi_4 - .003\xi_5 + .019\xi_6$$

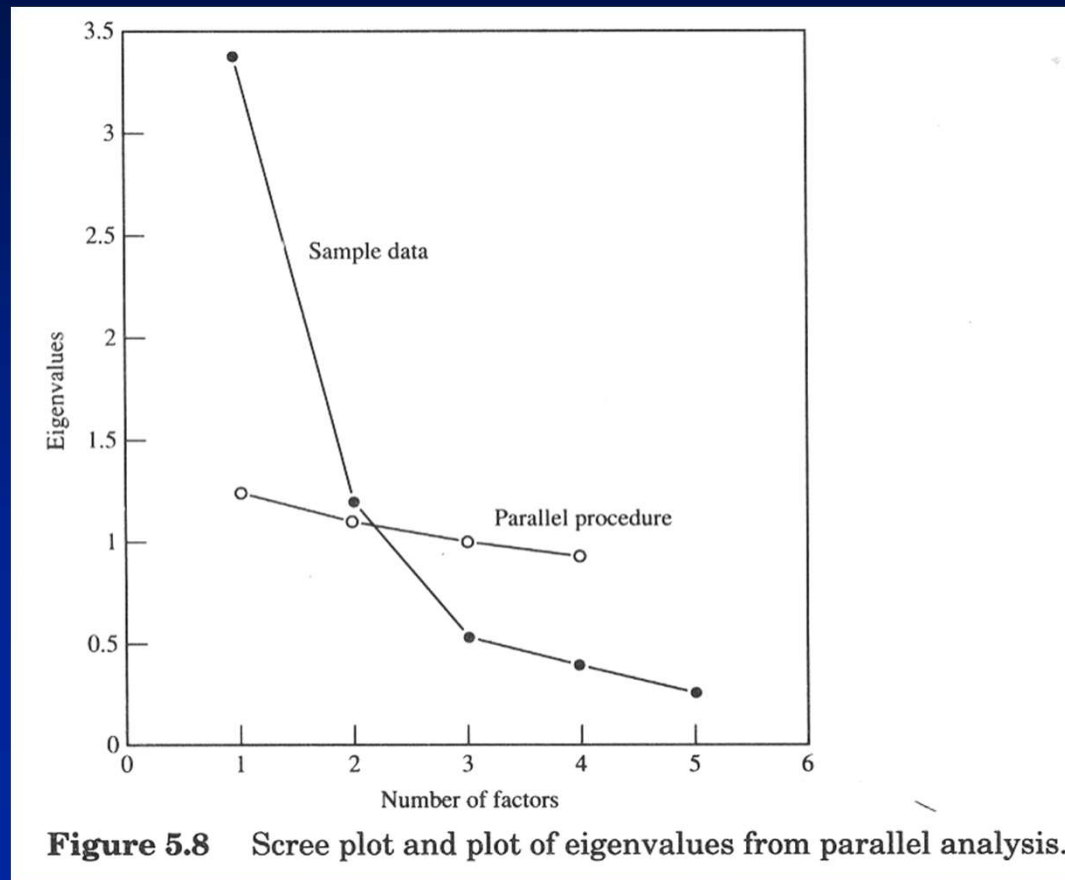
$$C = .683\xi_1 + .418\xi_2 + .592\xi_3 - .093\xi_4 - .002\xi_5 + .012\xi_6$$

$$E = .793\xi_1 - .410\xi_2 - .015\xi_3 + .040\xi_4 - .415\xi_5 + .171\xi_6$$

$$H = .774\xi_1 - .461\xi_2 - .002\xi_3 + .007\xi_4 + .373\xi_5 + .222\xi_6$$

$$F = .837\xi_1 - .359\xi_2 - .016\xi_3 + .021\xi_4 + .030\xi_5 - .409\xi_6.$$

Principal Components Factoring (PCF)



Principal Components Factoring (PCF)

Two factor FA model

$$M = .675\xi_1 + .557\xi_2 + \epsilon_m$$

$$P = .717\xi_1 + .447\xi_2 + \epsilon_p$$

$$C = .683\xi_1 + .418\xi_2 + \epsilon_c$$

$$E = .793\xi_1 - .410\xi_2 + \epsilon_e$$

$$H = .774\xi_1 - .461\xi_2 + \epsilon_h$$

$$F = .837\xi_1 - .359\xi_2 + \epsilon_f$$

Principal Components Factoring (PCF)

$$\begin{aligned}\epsilon_m &= -.190\xi_3 + .444\xi_4 + .027\xi_5 + .021\xi_6 \\ \epsilon_p &= -.346\xi_3 - .405\xi_4 - .003\xi_5 + .019\xi_6 \\ \epsilon_c &= .592\xi_3 - .093\xi_4 - .002\xi_5 + .012\xi_6 \\ \epsilon_e &= -.015\xi_3 + .040\xi_4 - .415\xi_5 + .171\xi_6 \\ \epsilon_h &= -.002\xi_3 + .007\xi_4 + .373\xi_5 + .222\xi_6 \\ \epsilon_f &= -.016\xi_3 + .021\xi_4 + .030\xi_5 - .409\xi_6.\end{aligned}$$

Principal Components Factoring (PCF)

Table 5.4 Summary of Principal Components Factor Analysis for the Correlation Matrix of Table 5.2

Variable	Factor Loadings		Communalities	Specific Variance ϵ
	ξ_1	ξ_2		
M	.675	.557	.766	.234
P	.717	.447	.714	.286
C	.683	.418	.641	.359
E	.793	-.410	.797	.203
H	.774	-.461	.812	.188
F	.837	-.359	.829	.171

Notes:

Principal Components Factoring (PCF)

Reproduced Correlation Matrix

	M	P	C	E	H	F
M	.766	.733	.694	.307	.266	.365
P	.733	.714	.677	.385	.349	.440
C	.694	.677	.641	.370	.336	.422
E	.307	.385	.370	.797	.803	.811
H	.266	.349	.336	.803	.812	.813
F	.365	.440	.422	.811	.813	.829

Note: Communalities are on the diagonal.

Reproduced Correlation Matrix

Principal Components Factoring

Residual correlation

	M	P	C	E	H	F
M	.234	-.113	-.154	.013	.018	.005
P	-.113	.285	-.167	-.005	.002	-.010
C	-.154	-.167	.359	-.010	.000	-.017
E	.013	-.005	-.010	.203	-.117	-.081
H	.018	.002	.000	-.117	.188	.079
F	.005	-.010	-.017	-.081	.078	.171

$$RMSR = \sqrt{\frac{\sum_{i=1}^p \sum_{j=i}^p res_{ij}^2}{p(p-1)/2}}$$

Principal Axis Factoring (PAF)

- An iterative procedure
 - Obtain loading pattern by performing PCA on original correlation matrix (Estimated communalities = 1).
 - Maximum change in estimated communalities is computed.
 - If the maximum change is large (?), then the original correlation matrix is modified by replacing the **diagonals** with the new estimated communalities.

Principal Axis Factoring (PAF)

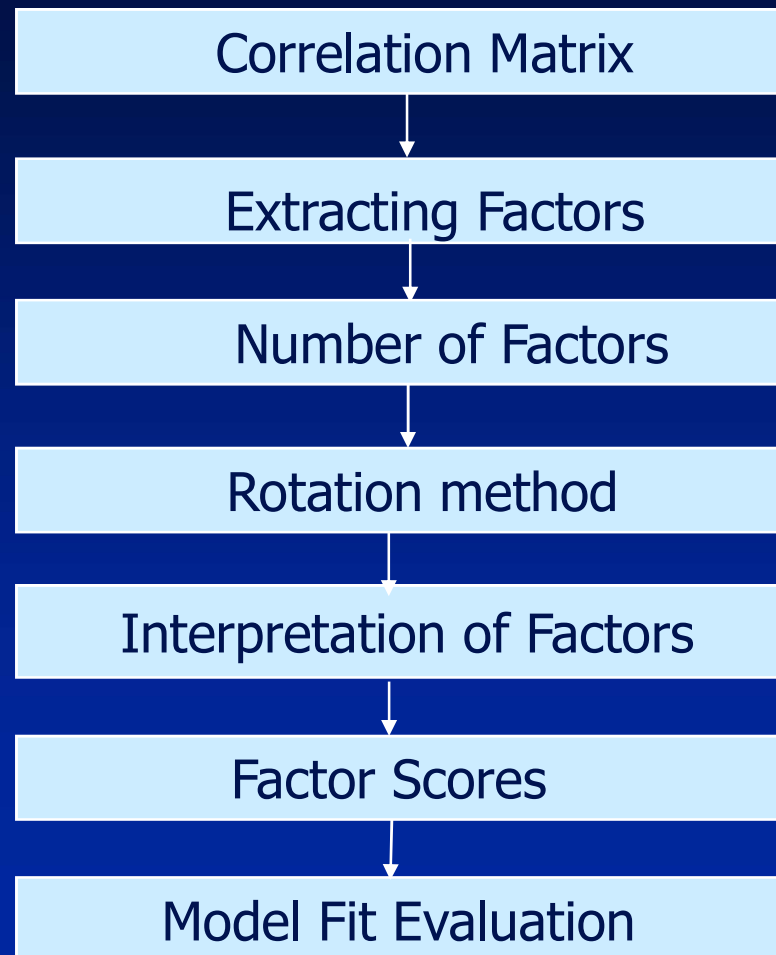
- The iteration will continue until it converge
 - maximum change < 0.001
 - 30 times of iteration

Principal Axis Factoring (PAF)

Iteration	Change	Communalities					
		M	P	C	E	H	F
1	.359	.766	.714	.641	.797	.812	.829
2	.128	.698	.626	.513	.725	.744	.784
3	.042	.679	.598	.471	.698	.719	.774
4	.014	.675	.588	.457	.688	.708	.774
5	.005	.674	.585	.453	.684	.703	.776
6	.003	.675	.583	.451	.682	.700	.779
7	.002	.676	.582	.451	.681	.698	.781
8	.001	.677	.582	.451	.681	.697	.782
9	.001	.677	.581	.450	.680	.697	.783

Iterative procedure for PAF

Procedure of Factor Analysis



Implementation of FA

- ❑ Data sets should have sufficient correlation and low partial correlation
- ❑ Is performed most often only on metric variables.
- ❑ The sample must have more observations than variables
- ❑ The minimum absolute sample size should be 50 observations.

Implementation of FA

- Group into homogenous classes
 - Examine correlation matrix
 - Examine partial correlation matrix
 - Kaiser's measure of overall sampling adequacy (MSA), which is based on above two matrixes and the sampling adequacy for each indicator (>0.5)

Implementation of FA

KMO Measure	Recommendation
$\geq .90$	Marvelous
.80+	Meritorious
.70+	Middling
.60+	Mediocre
.50+	Miserable
Below .50	Unacceptable

Implementation of FA_©

- The number of factors: using rules in PCA
- Model assessment
 - RMSR for overall and individual, like 0.001
- Interpretation
 - To achieve a simpler factor structure with meaningful interpretation.

Implementation of FA_©

- **Rotation**: to make the factor more understandable/interpretable
 - **Orthogonal** or oblique rotation
- **Orthogonal Rotation**
 - Varimax Rotation
 - Quartimax Rotation

Varimax Rotation

- Each factor will tend to have either large or small loadings of any particular variable.

$$\&_1 = -.156M - .063P - .030C + .303E + .346H + .454F$$

$$\&_2 = .534M + .339P + .215C - .045E - .092H - .029F,$$

Quartimax Rotation

- To obtain a pattern of loadings such all the variables have a fairly high loading on one factor (overall factor).
- Each variable should have a high loading on one same factor and near zero loadings on the remaining factors (specific construction).

SAS Code Example

```
DATA CORRMATR(TYPE=CORR);  
INPUT M P C E H F;  
-TYPE-='CORR';  
CARDS;  
Insert correlation matrix here  
PROC FACTOR  
METHOD =PRINIT  
ROTATE = V CORR MSA SCREE  
RESIDUALS PREPLOT PLOT;  
Run;
```


FA vs. PCA

- FA produces factors
- FA analyzes covariance (communality)
- FA is a close approximation to the R matrix
- The objective is to explain as much of the covariance with a minimum number of underlying factors
- FA can give multiple solutions depending on the method and the estimates of communality

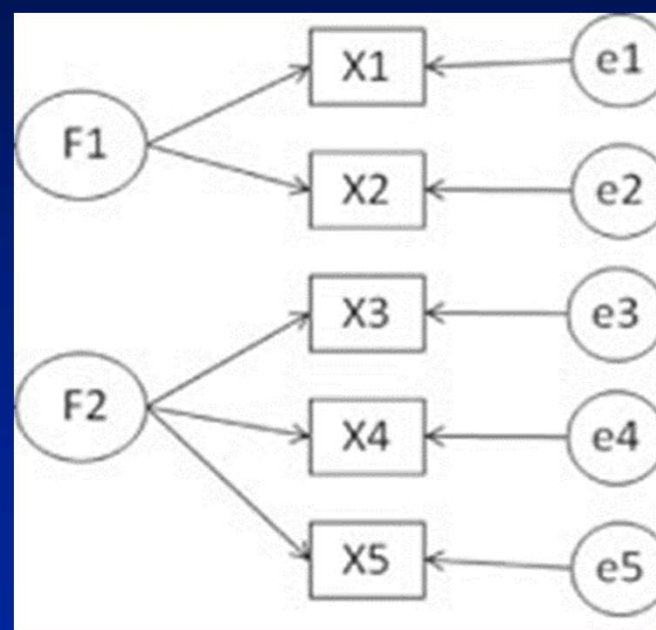
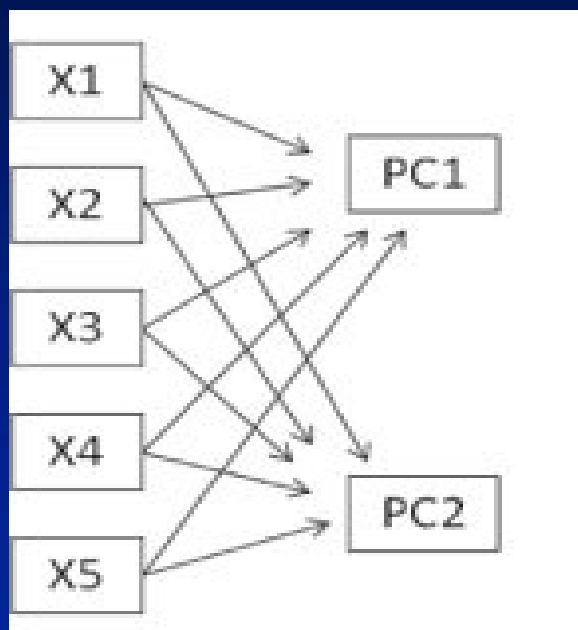
FA (vs. PCA)

- Three general goals: data reduction, describe relationships and explore theories about relationships
- How many interpretable factors exist in the data?
- How many factors are needed to summarize the pattern of correlations?

PCA (V.S. FA)

- The objective is to analyze variance and reduce the number of variables
- PCA decompose all variance into orthogonal components
- PCA reproduces the R matrix completely
- PCA extract as much variance with the least amount of factors
- PCA gives a unique solution

PCA V.S. FA



Questions

- ❑ Which factors account for the most variance?
- ❑ How well does the factor structure fit a given theory?
- ❑ What would each subject's score be if they could be measured directly on the factors?
- ❑ What does each factor mean?
- ❑ What is the percentage of variance in the data accounted for by the new variables?